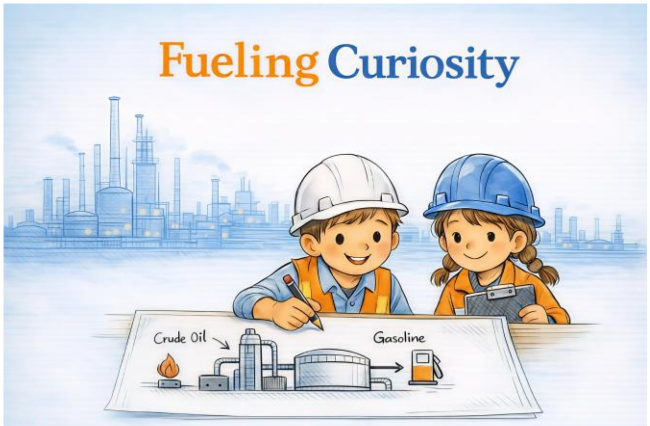

FUELING CURIOSITY STEM ACTIVITY PACK

Coloring Pages, Vocabulary, and Classroom Prompts

EXPLORE THE FASCINATING WORLD OF PETROLEUM REFINING THROUGH FUN, PRINTABLE ACTIVITIES.

This mini pack is designed for curious kids, parents, teachers, and STEM programs who want an accessible introduction to how refineries turn crude oil into useful everyday products.

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www.fuelingcuriosity.com

How to Use This Activity Pack

This printable pack is designed to help children explore the basics of petroleum refining in a simple, engaging way.

Ways to use it:

- Coloring activity for early learners
- STEM classroom warm-up
- Homeschool science enrichment
- Industry outreach and family events
- Discussion starter before or after reading Fueling Curiosity

Recommended ages:

- Coloring pages: ages 5+
- Vocabulary and discussion: ages 8+
- Extension activity: ages 8–12

Tip for teachers and parents:

Use the coloring pages first, then ask students what they notice. After that, use the vocabulary and discussion questions to connect the pictures to real engineering ideas.

Coloring Page: Fractionation Tower

A refinery separates crude oil into different useful products. Lighter products rise higher in the tower, while heavier products stay lower.

Think about it: Why do you think different liquids separate at different levels?

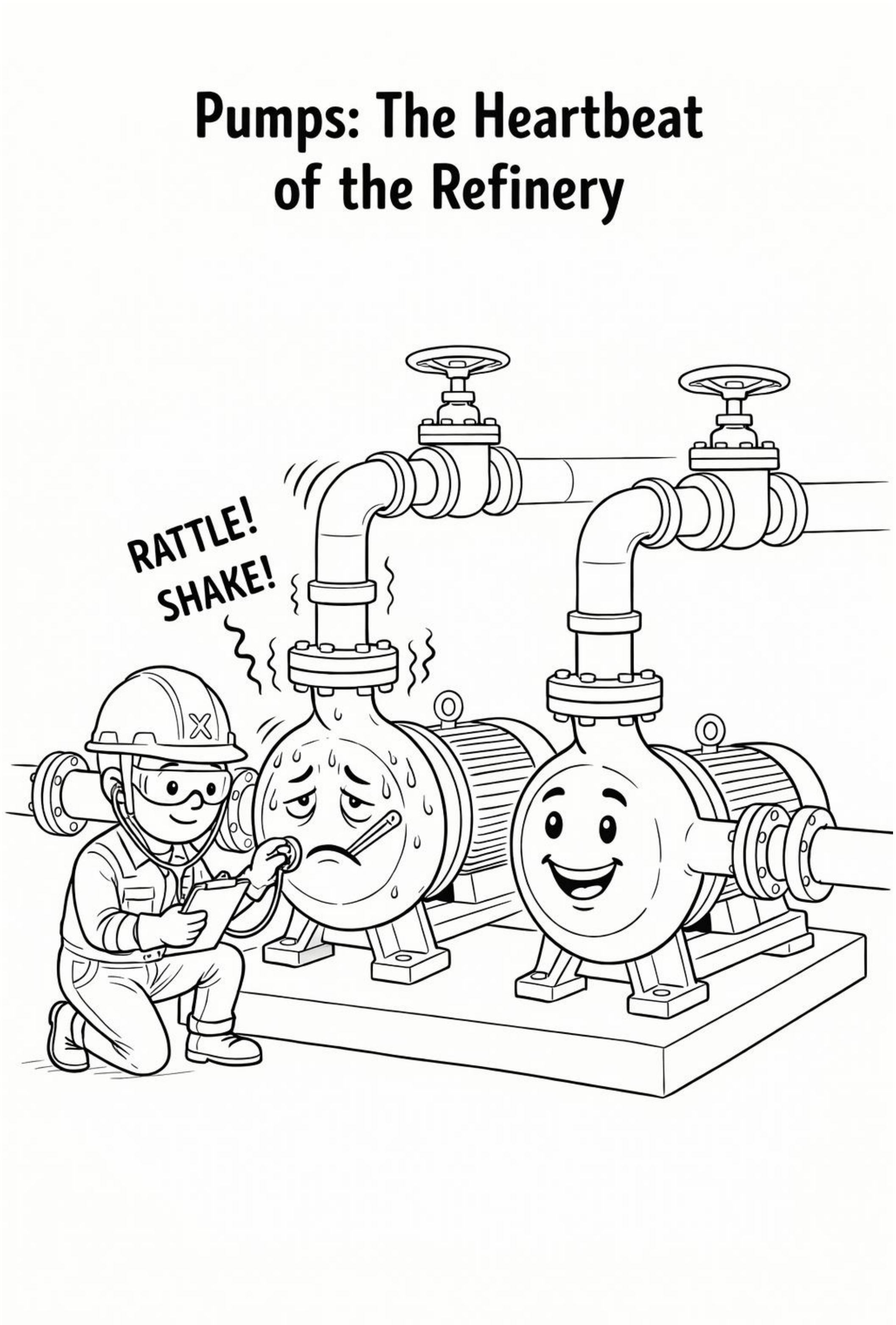
Coloring Page: Gasoline Blending

Making gasoline is not as simple as pouring one liquid into a tank. Refineries blend different components together to make fuels that meet important quality targets.

Think about it: Why might engineers test a fuel blend before sending it out to customers?

Coloring Page: Pumps and Reliability

Pumps help move liquids throughout a refinery. Keeping equipment running safely and reliably is an important part of refinery operations.



Think about it: What do you think happens if an important pump stops working?

Refinery Vocabulary

Vocabulary list

Crude Oil

A natural liquid found underground that can be turned into fuels and other useful products.

Refinery

A large industrial facility where crude oil is processed into gasoline, diesel, jet fuel, and more.

Fractionation Tower

A tall tower that separates heated crude oil into parts based on how easily they turn into gas.

Blend

To mix different materials together to make a final product with the right properties.

Gasoline

A blended fuel commonly used in cars and other small engines.

Diesel

A heavier fuel often used in trucks, trains, ships, and heavy equipment.

Jet Fuel

A fuel specially made for airplanes that burns cleanly and stays liquid at cold temperatures.

Pump

A machine that moves liquid from one place to another.

Engineer

A person who solves problems with math and science and helps design, operate, or improve systems and equipment.

Discussion Questions

1. What would happen if we did not have oil or refineries today?

2. Why do you think a refinery has many different pieces of equipment?

3. Why might fuel need to be tested before people use it?

4. What might happen if a refinery did not test or properly separate its products?

5. How do engineers help make fuels safer and more efficient for everyday use?

6. Why do different types of engines (like cars, airplanes, and ships) need different kinds of fuel?

7. What questions do you still have about how fuels are made or used?